

Caellum Yip Hoi-Lee

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EDUCATION

University of Toronto

Feb 2025 – Apr 2029

Honours Bachelor of Science — Bachelors of Computer Science

Toronto, ON

- **Cumulative GPA: 3.94/4.0**

EXPERIENCE

Founding Engineer

February 2026 – Present

Chatforce

Toronto, ON

- Overhauled Three.js agentic harness, implementing architecture, MCP, and tool calls for vision-driven multi-agent inspired by OpenGame/OpenCode; increased game publication rate by 20%.
- Built pixel art upscaler on Rust using fused-argmax voters, scored a 77% on pixel-bench.
- Developed FastAPI/Redis rate-limiting and server authoritative multiplayer, supporting **5000+ users**.
- Promoted to Founding Engineer within months after helping drive user count by over **900%**

Software Engineer/Project Lead, U of T Blueprint

Sep 2025 – May 2026

University of Toronto

Toronto, ON

- Leading a team of 10 developers to build software for 2 non-profit organizations.
- Developed JWT Authentication/Middleware, Admin Dashboard, and CSV Data Management system with Django REST and React for the Museum of Art and Digital Entertainment in California, streamlining staff workflows.

Owner and Software Lead

Aug 2023 – Present

Bronze Bat Studio

Richmond Hill, ON

- Directed an indie game dev studio, building mechanics, physics algorithms, and networking in Lua and C#.
- Published 7 interactive titles across Steam and Roblox, reaching over **3 million** total plays and **\$2,000** in revenue.

PROJECTS

Paper Cuts (Cal Hacks AI Hackathon | *Python, PyTorch, FastAPI, Redis, AWS Trainium, JavaScript* Jun 2026

- Won **1st Place Overall (\$5,000)** and **Best UI/UX** for a real-time game engine that converts user sketches into playable game assets, achieving sub-second latencies through fine tuning.
- Trained an SDXL diffusion model on AWS Trainium, a cascading CNN/VLM, and a Redis HNSW vector memory.
- Built “Moose,” a LoRA-tuned model mapping drawings to playable mechanics with latency under 500 ms.

Ensemble (Hack the Six | *Python, QNX Raspberry Pi, WebSockets, Qwen LoRA, Web Audio API* Jul 2026

- Won **Top 5 Overall** for an agentic conducting system that synchronizes connected devices via WebSockets.
- Used a QNX-based Raspberry Pi and an Arduino 4 wand to host a TinyML model and translate physical conductor gestures to live musical intent.
- Fine-tuned and deployed a real-time Qwen LoRA model on Fireworks H200 (453ms latency) that composes bars in real time; trained with SFT and GRPO on classical music theory.

Auctopus (SOON Hackathon) | *Next.js, Gemini 2.5 Pro, Veo 2, Neo4j, Cloudinary* May 2026

- Won **1st Place (Cloudinary & Polarity tracks)** for an automated AI product validation pipeline orchestrating Gemini 2.5 Pro, Veo 2, and Neo4j.
- Built a Next.js execution chain with SSE streaming, automating 3D AR model generation via the Meshy API and delivering them through Cloudinary’s CDN.
- Served enterprise analytics via Composio, utilized CyStack for secret injection and Polarity for product review.

SKILLS

Languages: Python, Java, C++, C#, JavaScript/TypeScript, SQL, Lua

Frameworks: React, Next.js, Django, FastAPI, React Native

AI & ML: PyTorch, LangChain, Hugging Face, RAG, Diffusion Models (SDXL)

Tools & Infrastructure: Linux/Unix, Git, Docker, AWS, PostgreSQL, Redis, Neo4j, WebSockets